

# A brief history of Agile and Lean



Learn. To Change.



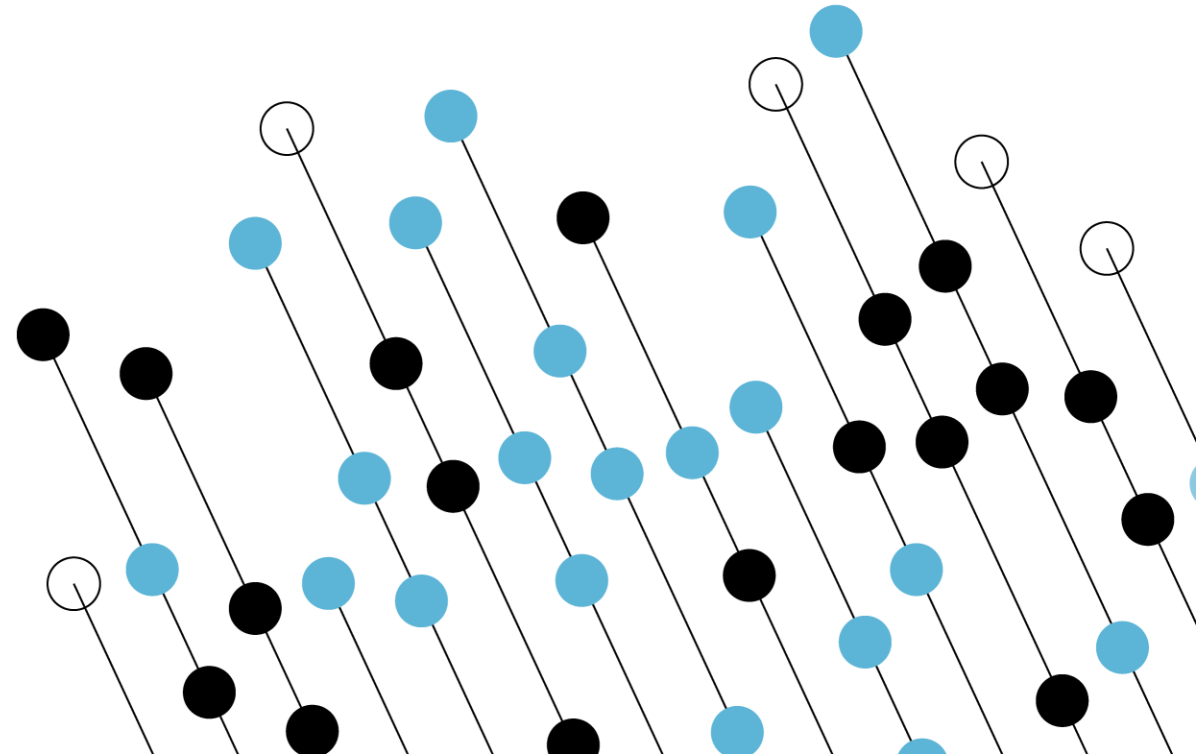
# Introduction

Agile delivery methods are being employed by more organisations as they try to extract increasing value from the work performed by their teams and deliver it earlier to their customer. They are characterised by their use of short periods of work and early releases of value-enabling outputs. For some environments, such as IT product and service delivery, these new ways of working have brought success; however, they are not suitable for all environments.

Most of these Agile approaches are created and implemented with only the delivery part of the organisation in mind. However, for true increased value and transformational change, the whole organisation must focus on becoming more agile.



# New ways of working



# New ways of working

The increasing pace of technological innovations from the mid-20th century onwards has had a profound effect on nearly all areas of our lives. The 'white heat of technology' spurred the rapid rise of ground-breaking products and services whose value was often poorly defined at the outset because the potential of the emergent technologies was not fully understood.

Out of this era emerged new ideas, models, and methods in manufacturing, engineering, and software design.



# Royce

Introduced by Dr Winston Royce of Lockheed Software Technology Center in his 1970 paper 'Managing the Development of Large Software Systems', the term 'Waterfall' refers to the cascading process structure (from start to finish) of the traditional delivery model.

Royce observed that, for many projects, the process discouraged communication between supplier and customer and led to duplication of effort when customers saw what had been created and asked for re-work. In fact, Royce commented that 'Whilst I believe in process, I believe it is risky and invites failure'.

Royce's work is accepted by many as the trigger for many new ideas and methods from both the manufacturing and software development fields which began to emerge and combine. Royce can therefore be viewed as the originator of agile ways of working.



# Takeuchi and Nonaka

Hirakata Takeuchi and Ikujiro Nonaka's famous article 'The New Kujira Product Development Game' brought together a range of ideas developed from Japanese manufacturing to champion a significantly different process model from Waterfall. This proved very influential for the development of Lean and Scrum approaches and well suited for new product development projects and start-ups.

They likened delivery to the game of rugby, and it is clear that this is the basis for Scrum. It is fair to say that without the work of Takeuchi and Nonaka, the Agile and Lean theories of today would have taken far longer to develop.

Their ideas featured six characteristics:

- Built-in stability
- Self-organising project teams
- Overlapping development phases
- Multi-learning
- Subtle control
- Organisational transfer of learning

To learn more about this work, visit:

<https://hbr.org/1986/01/the-new-new-product-development-game>

# Schwaber and Sutherland: Scrum

Ken Schwaber and Jeff Sutherland took the work created by Takeuchi and Nonaka and applied it to software development. First presented at a conference in 1995, over the following years Schwaber and Sutherland collaborated to develop what became known as Scrum. Since 2009 The Scrum Guide has been available as a free PDF download which is regularly updated.

Scrum is the most widely used Agile approach and is fully compatible with an AgileSHIFT environment.

To learn more about Scrum, Schwaber, and Sutherland, visit: <https://www.scrumguides.org/>



# Kanban method

Kanban is a term that covers the use of Kanban systems, which are visual work management systems that track the number of work items in circulation, therefore optimising the flow of items through a series of processes. This creates what is known as a 'Lean Pull System'. Kanban systems have for a long time existed in a wide variety of forms, for example, in the late 1940s Taiichi Ohno employed a system of signal cards to deliver the just-in-time element of the Toyota Production System.

The most common implementation of a Kanban system is the use of physical or virtual display boards (Kanban boards) that describe the steps of a process and the relative progress of deliverables through the process. This is a common work management tool in tech environments, as well as in many others.

Kanban is made up of six core practices:

- Visualise workflow
- Limit work in progress
- Measure and manage flow
- Make process policies explicit
- Use feedback loops
- Identify evolution opportunities

'Pure' Kanban has no iterations, while Kanban with iterations is sometimes known as 'Scrumban'.

# Lean and TPS

Lean is an accepted and convenient umbrella term for a number of techniques, practices, and processes that together enable organisations to better respond to consumer demand and market change. Lean is regarded by many manufacturing industries as the optimum operating model.

The Lean way of working was first formalised by the Toyota Motor Company of Japan through the creation and use of the Toyota Production System (TPS). Over the last 20 years, the term has been appropriated by many organisations and theorists with varying degrees of success. In some cases, it has been completely misunderstood and wrongly applied.

For organisations implementing AgileSHIFT, the most relevant parts of Lean will be the recognition of the importance of eliminating waste from the work that is being done and the role of the team in the achievement of outcomes valued by the customer.



# Lean Startup

Lean Startup is a methodology for guiding anyone who is starting a new product venture in the shortest route to delivery with the lowest exposure to risk. By removing waste, or working Lean, the leader of the start-up can work in a low-threat, high-impact way.

Significant characteristics of a Lean Startup environment include:

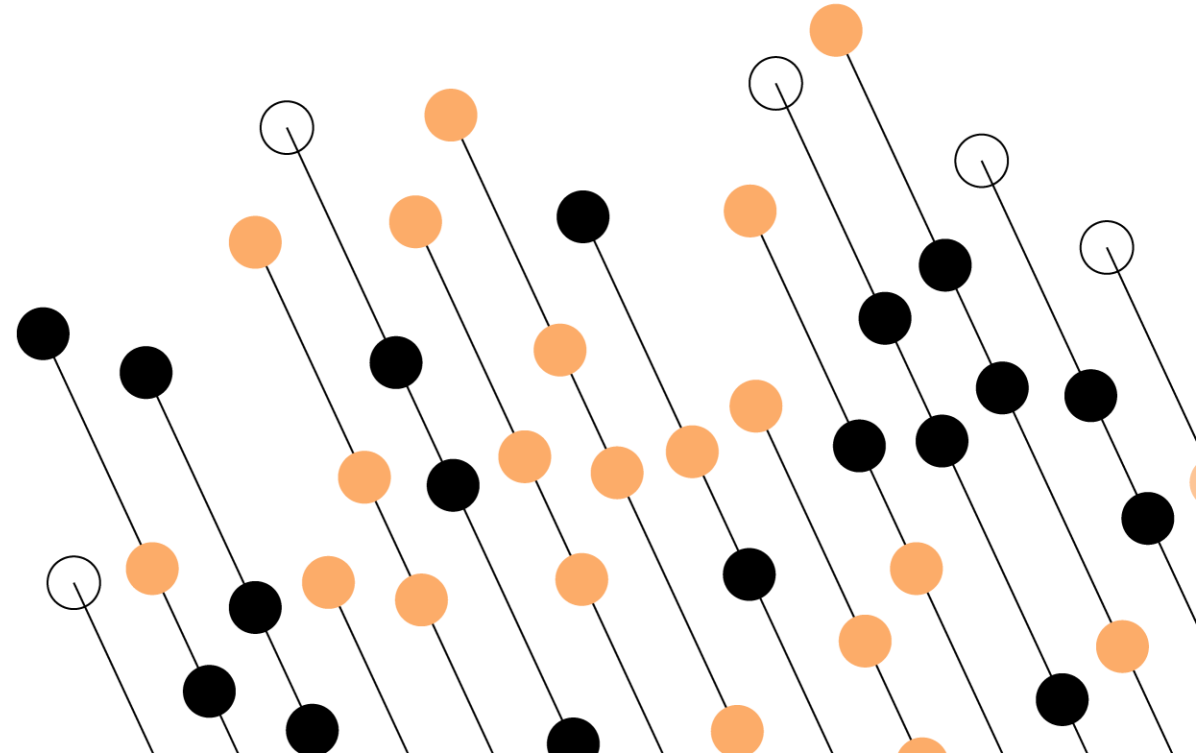
- Early release of a minimum viable product.
- Continuous deployment (of software).
- Validated learning (testing and measuring ideas).
- Innovation accounting (to maximise outcomes).
- Build-measure-learn (emphasising speed in product development).

To learn more about Lean Startup, visit:

<https://www.theleanstartup.com>

It is also worth noting that the author of The Lean Startup, Eric Ries, has published a further book, The Startup Way, this time encouraging large organisations to learn from successful start-ups.

# The rise of Agile and the Agile Manifesto



# The rise of Agile

In the early 1990s, software development faced a crisis in that the time between identifying a business need and the delivery of an actual application to meet that need was around three years. Within that period, requirements, technology, systems, and even entire organisations were likely to change.

As a result (and as a backlash against slow, unsuitable, 'heavyweight' processes), Agile software development processes were instigated. Since then, not only has there been a range of different Agile methods created, but the use of them has extended far beyond software development. Now, many teams in a wide range of sectors and industries would claim to be using Agile methods in some way.

The word Agile (with an upper case 'A') has come to be used as an umbrella term for new approaches to work that enable teams and individuals to work in a way that is typified by collaboration, prioritisation, and iterative and incremental delivery.



# The Agile Manifesto

The foundational movement for the coalescence of these practices into Agile was the publication of the 'Manifesto for Agile Software Development' in 2001. Created by 17 prominent software developers, the Agile Manifesto famously champions four key principles:

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

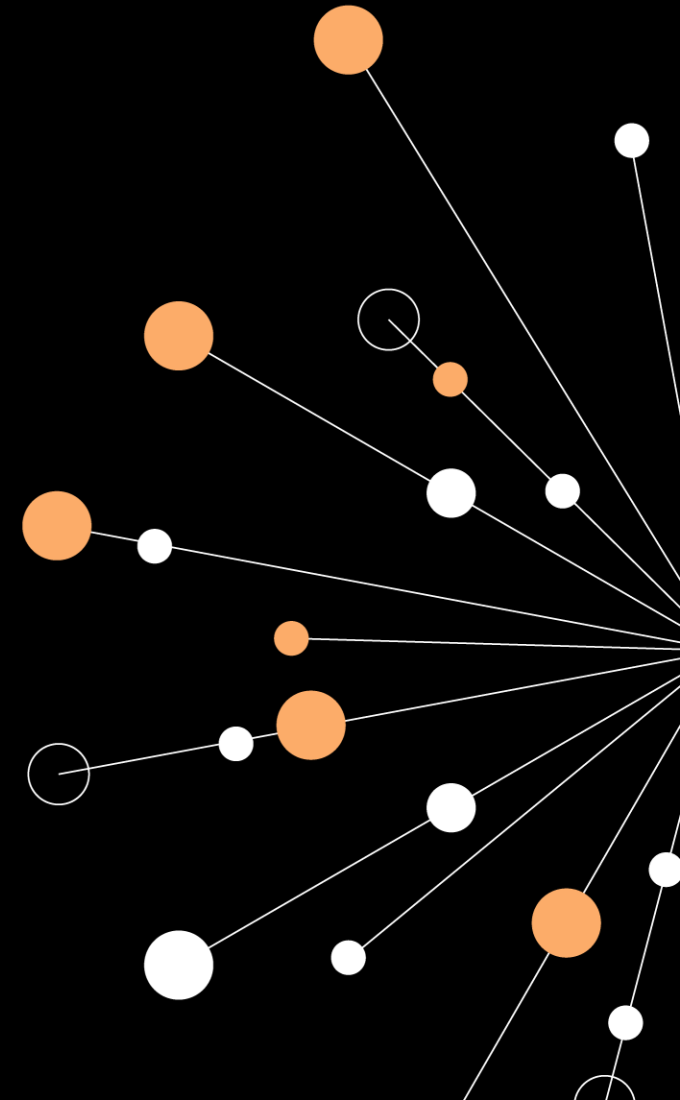
The authors state that 'while there is value in the items on the right, we choose to value the items on the left more'.



# Sector example: The need for Agile

In certain industries, the lag between identifying a need and delivery of a product was far greater than three years. In aerospace and defence, it could be 20 or more years before a complex system went into operation.

In an extreme but not unusual example, the Space Shuttle programme, which launched operationally in 1981, used information and processing technologies dating from the 1960s. Highly complex hardware and software systems were often designed, developed, and deployed in a timeframe that spanned decades.



# Popular Agile delivery approaches

Agile has now become an umbrella term for a range of incremental and iterative product and service delivery approaches and their associated tools. Examples of Agile delivery frameworks include:

- Scrum.
- Kanban.
- XP (Extreme Programming).
- SAFe (Scaled Agile Framework).
- Disciplined Agile Delivery.
- Lean.
- PRINCE™ Agile®.
- AgilePM®.

Agile delivery approaches are being employed by more organisations as they try to extract more value from the work performed by their teams. For some environments, these new ways of working have brought success; however, they are not suitable for all.

Most of these approaches were created and implemented with only the delivery part of the organisation in mind. But for true increased value and transformational change, the whole organisation must be involved. Its focus must be on becoming more responsive to develop an enterprise-wide culture of agility.